

Exercise I

AI Assistance Exercise

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

Today

- What is Python, what is an API, why APIs matter
- Your task, step by step

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- Email me when you are done, rli17@tulane.edu
- Completing the exercise earns the full credit (5% of the course grade)

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- The standard language for data and AI work, used across banks, funds, and tech firms
- An AI model writes the code for you today
- Your job is to read it, run it, and check the result

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- Same AI model behind both, the API is the door for programs

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- Real systems are built on APIs (trading systems, research pipelines, the apps on your phone)
- Today, one API call per firm on your list

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- You email me your results as soon as you are done
- I combine the class results into a portfolio of AI input firms and plot its return live

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- Built from the firm's official annual filings
- You get 30 firms, each with its full product profile
- The AI model reads the profiles and decides, tier by tier, whether the firm sells AI input products

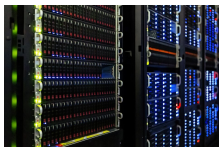
The AI Input List, Tier 1 (AI Core)

Item	Product category
T1_1	Data center AI accelerator chips
T1_2	High-bandwidth memory (HBM)
T1_3	AI-class networking and optical interconnects
T1_4	AI server systems and rack-scale designs
T1_5	GPU-dedicated AI cloud
T1_6	ML libraries, runtimes, and training platforms
T1_7	Foundation models and inference APIs



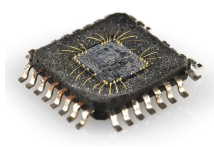
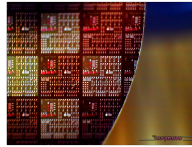
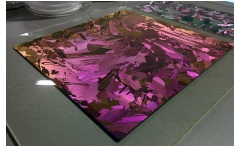
The AI Input List, Tier 2 (AI-Adjacent Infrastructure)

Item	Product category
T2_1	High-performance AI storage
T2_2	AI data center cooling
T2_3	Data center power distribution and backup
T2_4	Data center colocation and interconnection
T2_5	Data center construction and engineering
T2_6	Hyperscale cloud
T2_7	Edge AI silicon and modules



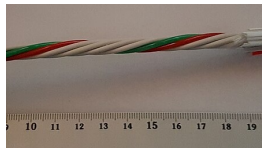
The AI Input List, Tier 3 (Semiconductor Manufacturing)

Item	Product category
T3_1	EDA software and chip design IP
T3_2	Fab and packaging materials
T3_3	Semiconductor capital equipment
T3_4	Contract wafer fabrication (foundries)
T3_5	Advanced chip packaging services



The AI Input List, Tier 4 (General Inputs)

Item	Product category
T4_1	Electric power generation and grid
T4_2	General industrial cooling and power equipment
T4_3	Commodity memory and storage
T4_4	Enterprise and telecom networking and optics



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5. Email me the results block the moment it appears

Working Time

- Notebook link is in my email (titled “FINE3010 Exercise I”)
- Your name goes into the prompt exactly as it appears on the roster
- Code errors? Paste the error back into ChatGPT and ask it to fix the code
- Email the results block to rli17@tulane.edu as soon as it appears, I plot the portfolio live
- No laptop or no Google account? Pair up with a neighbor and take turns
- Stuck? Raise your hand

Thank you!

Please let me know if you have any questions.